

Cancer-Associated Fibroblasts in Breast Cancer: From Heterogeneity to Therapeutic Targeting—A Systematic Review of Single-Cell and Spatial Omics Advances

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Abstract

Cancer-associated fibroblasts (CAFs) are a heterogeneous and dynamic population within the breast tumor microenvironment that critically influence tumor progression, metastasis, therapeutic resistance, and immune evasion. Traditionally viewed as passive structural support, CAFs are now recognised as active participants in tumor biology through paracrine signalling, extracellular matrix remodelling, and direct cell–cell interactions. The advent of single-cell RNA sequencing (scRNA-seq), spatial transcriptomics, and multi-modal omics technologies has revealed an unexpected degree of CAF heterogeneity, with distinct functional subpopulations co-existing within individual tumors. Here we provide a systematic review of CAF biology in breast cancer, organized into five interconnected themes: (I) the origins and activation mechanisms of CAFs, (II) single-cell-resolved CAF heterogeneity and subtyping, (III) spatial organization and niche-specific functions, (IV) CAF–immune cell crosstalk and its implications for immunotherapy, and (V) therapeutic strategies targeting CAFs. We survey more than 150 studies published between 2015 and 2026, integrate findings from bulk, single-cell, and spatial omics data, and propose a unified framework for understanding CAF functional diversity. We delineate the translational potential of CAF-targeted therapies, identify critical knowledge gaps, and discuss future directions including CAF-specific biomarkers, combination immunotherapy strategies, and the emerging role of CAFs in treatment resistance.

Keywords cancer-associated fibroblasts, breast cancer, tumor microenvironment, single-cell RNA sequencing, spatial transcriptomics, immunotherapy, therapeutic resistance

Introduction

Breast cancer remains the most commonly diagnosed malignancy and a leading cause of cancer-related mortality among women worldwide (Sung et al., 2021). Despite significant advances in early detection, surgical techniques, and systemic therapies, the heterogeneity of breast cancer—manifested at the molecular, cellular, and microenvironmental levels—continues to pose fundamental challenges to achieving durable responses and cures (Perou et al., 2000). The tumor microenvironment (TME), once regarded as a passive bystander, is now recognised as an active participant in tumor initiation, progression, metastasis, and therapeutic resistance (Hanahan and Weinberg, 2011).

Among the diverse cellular components of the TME, cancer-associated fibroblasts (CAFs) have emerged as a particularly abundant and functionally versatile population (Kalluri and LeBleu, 2016). In breast cancer, CAFs can constitute up to 80% of the tumor stroma, far exceeding their normal counterparts in healthy mammary tissue (Orimo et al., 2005). These activated fibroblasts differ from their quiescent normal counterparts in morphology, gene expression, and functional capabilities, actively promoting tumor progression

through multiple mechanisms including extracellular matrix (ECM) remodelling, paracrine signalling, metabolic reprogramming, and immune modulation (Sahai et al., 2020).

Historically, CAFs were viewed as a relatively homogeneous population of tumor-supportive cells. However, the application of single-cell RNA sequencing (scRNA-seq) and spatial transcriptomics technologies has fundamentally challenged this view, revealing an unexpected degree of heterogeneity within the CAF compartment (Bartoschek et al., 2018; Lambrechts et al., 2018). Multiple transcriptionally distinct CAF subpopulations—including myfibroblastic CAFs (myCAFs), inflammatory CAFs (iCAFs), and antigen-presenting CAFs (apCAFs)—have been identified across breast cancer subtypes, each with distinct spatial distributions, functional roles, and clinical implications (Erve et al., 2024).

This heterogeneity has profound implications for both fundamental biology and clinical practice. Different CAF subpopulations may promote or restrain tumor growth, enhance or suppress immune responses, and confer sensitivity or resistance to specific therapies (Chhabra et al., 2023). Understanding the full spectrum of CAF diversity—its origins, spatial organization, functional specialization,

and therapeutic vulnerabilities—is therefore essential for developing effective CAF-targeted treatment strategies.

Existing reviews have addressed various aspects of CAF biology (Kalluri and LeBleu, 2016; Sahai et al., 2020; Lavie et al., 2022), but none has systematically integrated findings from the rapidly accumulating single-cell and spatial omics datasets specific to breast cancer. Here we fill this gap by providing a comprehensive review organized around five interconnected themes:

1. **CAF Origins and Activation:** the cellular sources of CAFs and the signalling pathways governing their activation (Section 2).
2. **Single-Cell Heterogeneity:** the transcriptionally and functionally distinct CAF subpopulations revealed by scRNA-seq (Section 3).
3. **Spatial Organization:** how spatial transcriptomics and imaging technologies map CAF subtypes to specific tissue niches (Section 4).
4. **CAF–Immune Crosstalk:** the bidirectional signalling between CAFs and immune cells, and its implications for immunotherapy (Section 5).
5. **Therapeutic Targeting:** current and emerging strategies for eliminating or reprogramming CAFs (Section 6).

We further discuss CAFs as mediators of treatment resistance (Section 7), their potential as biomarkers (Section 8), and identify critical gaps and future directions (Section 9). By integrating evidence across experimental models, clinical datasets, and computational analyses, we aim to provide a unified framework for understanding CAF biology in breast cancer and to guide the development of CAF-targeted therapies.

CAF Origins and Activation Mechanisms

The cellular origins of CAFs in breast cancer are more diverse than previously appreciated. While tissue-resident fibroblasts were long considered the primary source, accumulating evidence indicates that CAFs can arise from multiple precursor populations through distinct activation pathways.

Tissue-resident fibroblasts as the predominant source

In the normal mammary gland, fibroblasts exist in a quiescent state, maintaining tissue homeostasis through controlled ECM production and paracrine signalling with epithelial cells (Plaks et al., 2015). Upon malignant transformation, these resident fibroblasts become activated through exposure to tumour-derived factors, most notably transforming growth factor- β (TGF- β), platelet-derived growth factor (PDGF), and fibroblast growth factors (FGFs) (Kalluri and LeBleu, 2016).

TGF- β signalling represents the best-characterised activation pathway. Tumour cells and immune cells secrete latent TGF- β , which is activated in the ECM through integrin-mediated mechanical force or proteolytic cleavage (Shi et al., 2011). Active TGF- β binds to TGF- β receptors on fibroblasts, triggering SMAD2/3 phosphorylation and nuclear translocation, which drives the transcription of myofibroblast markers including α -smooth muscle actin (α -SMA), collagen type I, and fibronectin (Derynck and Zhang, 2001). This TGF- β -driven activation programme is particularly prominent in myCAF, the most abundant CAF subpopulation in breast tumours (Erve et al., 2024).

PDGF signalling operates in parallel, with tumour-secreted PDGF-AA and PDGF-BB activating PDGF receptor α (PDGFR α) and

PDGFR β on fibroblasts, respectively (Ostman and Heldin, 2001). PDGF signalling promotes fibroblast proliferation, migration, and ECM production, and is essential for establishing the desmoplastic reaction characteristic of many breast cancers (Shao et al., 2016).

Bone marrow-derived mesenchymal stem cells

A proportion of CAFs originate from bone marrow-derived mesenchymal stem cells (BM-MSCs) that are recruited to the tumour site through chemokine gradients (Spaeth et al., 2009). CXCL12 (SDF-1) secreted by tumour cells and activated stroma attracts CXCR4-expressing BM-MSCs, which subsequently differentiate into CAFs under the influence of tumour-derived TGF- β and IL-6 (Karnoub et al., 2007).

In breast cancer, BM-MSC-derived CAFs have been shown to promote tumour growth, angiogenesis, and metastasis more potently than their tissue-resident counterparts (Quante et al., 2011). This functional superiority may reflect the acquisition of epigenetic modifications during bone marrow residence that prime these cells for rapid activation upon tumour recruitment. However, the relative contribution of BM-MSCs to the total CAF pool varies across breast cancer subtypes and individual tumours, with estimates ranging from 5% to 25% (Augello and De Bari, 2005).

Endothelial-to-mesenchymal transition

Endothelial cells can acquire fibroblast-like properties through endothelial-to-mesenchymal transition (EndMT), a process driven by TGF- β , Wnt, and Notch signalling (Zeisberg et al., 2007). During EndMT, endothelial cells downregulate vascular endothelial cadherin (VE-cadherin) and CD31 while upregulating α -SMA, fibroblast-specific protein 1 (FSP1), and collagen production (Potenta et al., 2008).

In breast cancer, EndMT-derived CAFs have been identified in the perivascular niche, where they contribute to vascular normalization and immune cell trafficking (Gasparri et al., 1999). These cells may represent a distinct CAF subpopulation with specialized functions in angiogenesis regulation and immune cell extravasation, though their precise contribution to the overall CAF landscape remains to be fully characterized.

Epithelial-to-mesenchymal transition as a source

Epithelial-to-mesenchymal transition (EMT) can generate CAF-like cells from malignant epithelial cells themselves (Petersen et al., 2003). Through EMT, tumour cells lose epithelial markers (E-cadherin, cytokeratins) and acquire mesenchymal properties (N-cadherin, vimentin, α -SMA), becoming indistinguishable from fibroblasts by standard markers (Kalluri and Weinberg, 2009).

This phenomenon—termed “tumour-cell-derived CAFs”—is particularly relevant in breast cancer, where EMT is a central mechanism of metastasis and therapeutic resistance (Niegel et al., 2018). However, distinguishing tumour-derived CAFs from bona fide fibroblast-derived CAFs requires lineage tracing or tumour-specific genetic markers, which are rarely available in clinical samples. The functional significance of tumour-derived CAFs remains debated: some studies suggest they are more aggressive than conventional CAFs (Chen et al., 2014), while others argue they represent a minor population with limited impact on tumour biology (Strell et al., 2019).

Activation signalling networks

Beyond the canonical pathways described above, CAF activation in breast cancer involves a complex network of interacting signals:

IL-6/STAT3 axis: Tumour-derived IL-6 activates STAT3 signalling in fibroblasts, promoting their differentiation into iCAFs characterized by high expression of inflammatory cytokines (IL-6, IL-8, CXCL1) and chemokines (CCL2, CCL5) (Erve et al., 2024). This axis creates a feed-forward loop: iCAF-secreted IL-6 further activates STAT3 in tumour cells, enhancing their proliferative and invasive capacity.

Hedgehog signalling: Tumour-secreted Sonic hedgehog (SHH) activates the Hedgehog pathway in fibroblasts through the receptor PTCH1 and effector GLI2, driving ECM production and desmoplasia (Rao et al., 2006). In breast cancer, Hedgehog signalling is particularly active in the basal-like subtype, where it contributes to the characteristic dense stroma.

Notch signalling: Direct cell–cell contact between tumour cells and fibroblasts activates Notch signalling in the latter, promoting their proliferation and differentiation (Strell et al., 2019). This juxtacrine mechanism operates independently of secreted factors and may be particularly important at the tumour–stroma interface.

Mechanical signalling: The stiff, collagen-rich ECM produced by activated fibroblasts generates mechanical forces that are sensed by mechanosensitive ion channels and integrins on fibroblasts themselves, reinforcing their activated phenotype (Calvo et al., 2013). This mechanical feedback loop contributes to the self-sustaining nature of the CAF phenotype once established.

Heterogeneity of activation states

Importantly, fibroblast activation is not a binary switch but a continuum of states. Single-cell studies have revealed that CAFs within a single tumour occupy multiple activation states, ranging from quiescent to fully activated (Erve et al., 2024; Bartoschek et al., 2018). The balance between these states is influenced by local signalling gradients, ECM composition, and oxygen tension.

This heterogeneity of activation states has important implications for therapeutic targeting: interventions that block a single activation pathway may shift the equilibrium toward alternative states rather than eliminating CAFs entirely. A systems-level understanding of the activation network is therefore essential for designing effective CAF-targeted strategies.

Single-Cell Resolution of CAF Heterogeneity

The application of single-cell RNA sequencing to breast tumours has transformed our understanding of CAF biology, revealing a degree of heterogeneity that was invisible to bulk transcriptomic analyses. This section synthesises findings from more than 30 scRNA-seq studies published between 2018 and 2026, integrating them into a unified framework of CAF subtyping.

Discovery of major CAF subpopulations

The first single-cell atlas of breast cancer CAFs was reported by Bartoschek et al. (2018), who identified three transcriptionally distinct fibroblast populations in mouse mammary tumours: a TGF- β -responsive population, a Hedgehog-responsive population, and a quiescent population. This foundational work established that CAF

heterogeneity is not random noise but reflects stable, functionally specialized subpopulations.

Subsequent studies in human breast cancer confirmed and extended these findings. Erve et al. (2024) performed a comprehensive meta-analysis of scRNA-seq data from 26 breast tumours, identifying three major CAF subpopulations that have become the reference classification:

- **myCAFs** (myofibroblastic CAFs): characterized by high expression of α -SMA (*ACTA2*), collagen genes (*COL1A1*, *COL3A1*), and fibronectin (*FN1*). myCAFs are the most abundant subpopulation, typically comprising 40–60% of CAFs in breast tumours. They are localized primarily in the immediate peritumoural stroma and are responsible for ECM production and tissue stiffness.
- **iCAFs** (inflammatory CAFs): defined by expression of inflammatory cytokines (*IL6*, *IL8*, *CXCL1*) and chemokines (*CCL2*, *CCL5*). iCAFs are enriched at the tumour margin and in areas of immune cell infiltration. They lack α -SMA expression, distinguishing them from myCAFs.
- **apCAFs** (antigen-presenting CAFs): expressing MHC class II molecules (*HLA-DRA*, *CD74*) and co-stimulatory molecules (*CD80*, *CD86*). apCAFs are the smallest subpopulation (10–20% of CAFs) and are found predominantly at the tumour–immune interface.

This tripartite classification, while useful, represents a simplification of the true CAF landscape. Higher-resolution analyses have revealed additional subpopulations and continuous transcriptional gradients within each major class (Wu et al., 2020).

Functional specialization of CAF subpopulations

Each major CAF subpopulation performs distinct functions that collectively support tumour progression:

myCAFs: ECM architects and mechanical regulators

myCAFs are the primary producers of the dense, collagen-rich ECM that characterizes breast tumour stroma (Erve et al., 2024). Beyond structural support, this ECM serves as a reservoir for growth factors (TGF- β , VEGF, FGF) that are released during ECM remodelling, creating a feed-forward loop that sustains CAF activation (Lu et al., 2012).

The mechanical properties of the myCAF-produced ECM have profound biological consequences. Increased tissue stiffness activates mechanosensitive pathways in both tumour cells and CAFs themselves, promoting proliferation, invasion, and resistance to apoptosis (Levental et al., 2009). In breast cancer, high stromal stiffness—largely attributable to myCAF activity—is an independent predictor of poor prognosis and correlates with lymph node metastasis (Levental et al., 2009).

myCAFs also contribute to tumour invasion through physical remodelling of the ECM. By aligning collagen fibres perpendicular to the tumour border (termed “tumour-associated collagen signatures” or TACS), myCAFs create physical tracks that facilitate tumour cell migration (Provenzano et al., 2006). This collagen alignment is mediated by myCAF contractility through Rho/ROCK signalling and can be disrupted by pharmacological inhibition of this pathway.

iCAFs: paracrine signalling hubs

iCAFs function as paracrine signalling centres, secreting a wide range of cytokines and chemokines that modulate both tumour cells and immune cells (Erve et al., 2024). The iCAF secretome includes:

- **IL-6:** activates STAT3 signalling in tumour cells, promoting proliferation, survival, and EMT. IL-6 also drives the differentiation of CD4⁺ T cells into Th17 cells and inhibits regulatory T cell (Treg) suppression (Chang et al., 2014).
- **CXCL1/CXCL8:** attract neutrophils and myeloid-derived suppressor cells (MDSCs) to the tumour site, contributing to immune suppression (Acharyya et al., 2012).
- **CCL2:** recruits monocytes and macrophages, promoting the establishment of a tumour-promoting myeloid compartment (Qian and Pollard, 2011).
- **CXCL12:** attracts CXCR4⁺ endothelial progenitor cells, supporting angiogenesis, and recruits Tregs, enhancing immune evasion (Orimo et al., 2005).

The iCAF phenotype is driven by NF- κ B signalling, which is activated by tumour-derived TNF- α , IL-1 β , and pathogen-associated molecular patterns (PAMPs) from the gut microbiome (Erve et al., 2024). This inflammatory programme is self-reinforcing: iCAF-secreted cytokines activate NF- κ B in neighbouring fibroblasts, expanding the iCAF compartment.

apCAFs: immune modulators

apCAFs represent the most recently characterized subpopulation and remain the least understood. These cells express MHC class II molecules, enabling them to present antigens to CD4⁺ T cells, but lack classical co-stimulatory molecules (CD80/CD86), suggesting a tolerogenic rather than immunogenic function (Erve et al., 2024).

In breast cancer, apCAFs have been shown to induce T cell anergy and promote immune evasion through antigen presentation without co-stimulation (Pinchuk et al., 2008). This mechanism is analogous to the tolerogenic function of plasmacytoid dendritic cells in the tumour microenvironment. However, the relative importance of apCAFs versus professional antigen-presenting cells (dendritic cells, macrophages) in shaping the anti-tumour immune response remains to be determined.

Context-dependent CAF heterogeneity

CAF subpopulation composition varies substantially across breast cancer subtypes, suggesting that CAF heterogeneity is shaped by the molecular characteristics of the tumour cells:

ER⁺ breast cancer: Characterized by a predominance of myCAFs with a well-organized, collagen-rich stroma. iCAFs are present but less abundant, and apCAFs are relatively rare (Wu et al., 2020). The desmoplastic reaction in ER⁺ tumours is particularly pronounced, contributing to the characteristic “hard” palpable mass.

HER2⁺ breast cancer: Contains a higher proportion of iCAFs compared to ER⁺ tumours, with increased expression of inflammatory cytokines and chemokines. This inflammatory CAF signature correlates with enhanced immune infiltration and may contribute to the efficacy of HER2-targeted immunotherapies (Liu et al., 2019).

Triple-negative breast cancer (TNBC): Displays the most heterogeneous CAF landscape, with variable proportions of all three major subpopulations. TNBC tumours often contain a subset of CAFs

with stem-like properties (expressing *CD44*, *ALDH1*), which may contribute to treatment resistance and recurrence (Peer et al., 2021).

Basal-like breast cancer: Enriched for myCAFs with high Hedgehog signalling activity and a particularly dense, stiff stroma. The CAF compartment in basal-like tumours is more homogeneous than in TNBC, suggesting a dominant activation pathway (Peinado et al., 2020).

Temporal dynamics of CAF heterogeneity

CAF subpopulations are not static but evolve during tumour progression and in response to therapy. Longitudinal scRNA-seq studies in mouse models have revealed:

- **Early tumours:** dominated by quiescent fibroblasts with a minor iCAF population, suggesting that inflammatory activation precedes myofibroblast differentiation.
- **Advanced tumours:** progressive enrichment of myCAFs with concurrent loss of quiescent fibroblasts, consistent with the increasing desmoplasia observed clinically.
- **Post-therapy:** rapid expansion of iCAFs following chemotherapy or immunotherapy, potentially contributing to treatment resistance through inflammatory signalling (Peer et al., 2021).

These temporal dynamics suggest that the optimal therapeutic strategy for targeting CAFs may differ at different stages of disease and that interventions must account for the adaptive capacity of the CAF compartment.

Cross-species conservation of CAF heterogeneity

A critical question for translational research is the extent to which CAF subpopulations identified in mouse models correspond to those in human tumours. Comparative analyses have revealed substantial but incomplete conservation:

Conserved features: The myCAF/iCAF/apCAF classification is broadly conserved across species, with orthologous gene expression programmes and similar spatial distributions (Erve et al., 2024).

Species-specific differences: Human CAFs display greater transcriptional diversity than their mouse counterparts, with additional subpopulations that have no clear mouse equivalent. This species-specific heterogeneity may reflect differences in tumour aetiology, immune system organization, or stromal biology.

These findings highlight the importance of validating CAF-targeted therapies in human systems and the need for caution when extrapolating from mouse models to clinical practice.

Spatial Organization of CAFs in Breast Tumours

The spatial distribution of CAF subpopulations within breast tumours is not random but reflects organized niches with distinct functional roles. Spatial transcriptomics and multiplexed imaging technologies have begun to map this spatial architecture, revealing how CAF positioning influences tumour biology.

Spatial transcriptomics reveals CAF niches

Spatial transcriptomics platforms (Visium, MERFISH, Xenium) enable simultaneous measurement of gene expression and spatial coordinates, allowing CAF subpopulations to be mapped to specific tissue regions (Ståhl et al., 2016).

Wu et al. (2020) combined scRNA-seq with Visium spatial transcriptomics to map CAF distributions in breast tumours, identifying three distinct spatial niches:

- **Peritumoural niche:** immediately adjacent to tumour cells, dominated by myCAFs expressing high levels of collagen and α -SMA. This niche is characterized by stiff, aligned collagen fibres that facilitate tumour cell invasion.
- **Invasive front:** at the tumour–stroma boundary, enriched for iCAFs and apCAFs. This niche is the primary site of tumour–immune cell interaction and contains the highest density of infiltrating lymphocytes.
- **Distant stroma:** well away from tumour cells, containing quiescent fibroblasts and a small population of iCAFs. This niche may serve as a reservoir for CAF precursors that are recruited to the tumour as it expands.

This spatial organization is not static but evolves during tumour progression. Longitudinal spatial analyses in mouse models have shown that the peritumoural niche expands as tumours grow, with progressive recruitment of iCAFs to the invasive front (Erve et al., 2024).

Spatial co-localization with immune cells

Spatial transcriptomics has revealed that CAF subpopulations are not uniformly distributed relative to immune cells but show specific co-localization patterns:

iCAFs and macrophages: iCAFs are frequently co-localized with tumour-associated macrophages (TAMs), particularly in the invasive front niche. This spatial proximity facilitates paracrine signalling: iCAF-secreted CCL2 recruits monocytes that differentiate into TAMs, while TAM-secreted TGF- β maintains iCAF activation (Qian and Pollard, 2011).

apCAFs and T cells: apCAFs show the strongest spatial association with CD4⁺ T cells, consistent with their antigen-presenting function. Single-cell spatial analyses have revealed direct cell–cell contacts between apCAFs and T cells at the tumour–immune interface, suggesting active antigen presentation (Erve et al., 2024).

myCAFs and tumour cells: myCAFs form a physical barrier around tumour cell clusters, creating a “fibrotic capsule” that may limit immune cell infiltration. This spatial arrangement has been termed “immune exclusion” and is associated with resistance to immunotherapy (Chhabra et al., 2023).

Imaging-based spatial proteomics

Beyond transcriptomics, imaging-based technologies (CODEX, IMC, MIBI-TOF) enable simultaneous measurement of 40+ proteins at single-cell resolution, providing complementary spatial information to transcriptomic approaches (Goltsev et al., 2018).

Keren et al. (2018) used multiplexed ion beam imaging (MIBI) to map the spatial proteome of breast tumours, revealing that CAFs expressing podoplanin (PDPN) and α -SMA form distinct spatial clusters with different immune cell compositions. PDPN⁺ CAFs were preferentially located near CD8⁺ T cells, while α -SMA⁺ CAFs were associated with macrophages.

These imaging-based approaches provide spatial information at subcellular resolution that is complementary to the whole-transcriptome coverage of spatial transcriptomics. Integration of both modalities offers a comprehensive view of CAF spatial biology,

though technical challenges in data alignment and cross-platform normalization remain.

Spatial gradients and niche boundaries

Spatial transcriptomics has revealed that CAF subpopulations are not abruptly demarcated but form gradients across tissue regions. This is particularly evident for the myCAF–iCAF transition, which occurs over a distance of 100–200 μ m from the tumour border (Erve et al., 2024).

These spatial gradients likely reflect diffusion gradients of tumour-derived signals (TGF- β , IL-6, TNF- α) that drive differential CAF activation. The steepness of these gradients varies across tumours and is influenced by vascular density, ECM composition, and interstitial fluid pressure.

Understanding spatial gradients has important implications for therapy: interventions that alter the signalling landscape (e.g., anti-TGF- β therapy) may reshape CAF spatial organization, potentially converting an immunosuppressive niche into an immunostimulatory one.

Spatial transcriptomics-guided CAF subtyping

The integration of scRNA-seq with spatial transcriptomics has enabled spatially resolved CAF subtyping that goes beyond the classical myCAF/iCAF/apCAF classification. Erve et al. (2024) identified six spatially distinct CAF states in breast tumours, each with a unique spatial niche and functional programme:

1. **ECM-myCAFs:** located in the peritumoural niche, producing dense collagen matrices.
2. **Contractile myCAFs:** found at the invasive front, generating mechanical forces for tissue remodelling.
3. **Inflammatory iCAFs:** distributed throughout the stroma, producing cytokines and chemokines.
4. **Regulatory iCAFs:** enriched near immune aggregates, expressing immune checkpoint ligands.
5. **apCAFs:** localized at the tumour–immune interface, presenting antigens to T cells.
6. **Quiescent fibroblasts:** found in distant stroma, maintaining tissue homeostasis.

This spatially resolved subtyping provides a more nuanced understanding of CAF biology and may enable more precise therapeutic targeting of specific CAF subpopulations in defined spatial contexts.

Clinical implications of CAF spatial organization

The spatial distribution of CAFs has direct clinical relevance:

Prognostic significance: High density of iCAFs at the invasive front is associated with poor prognosis in breast cancer, reflecting the pro-tumorigenic and immunosuppressive functions of these cells (Erve et al., 2024). Conversely, high density of quiescent fibroblasts in the distant stroma is associated with better outcomes, suggesting a protective role.

Predictive value: The spatial organization of CAFs may predict response to immunotherapy. Tumours with dense myCAF capsules surrounding tumour cell clusters (immune-excluded phenotype) show poor response to immune checkpoint inhibitors, while tumours

with dispersed iCAFs and apCAFs (immune-infiltrated phenotype) respond better (Chhabra et al., 2023).

Therapeutic implications: Understanding CAF spatial organization may guide the design of CAF-targeted therapies. For example, therapies that disrupt the myCAF capsule may enhance immune cell infiltration, while therapies that reprogram iCAFs may reduce immunosuppression.

CAF–Immune Cell Crosstalk in Breast Cancer

The bidirectional communication between CAFs and immune cells is a central feature of the breast tumour microenvironment. CAFs shape the immune landscape through multiple mechanisms, while immune cells reciprocally regulate CAF activation and function. This crosstalk has profound implications for immunotherapy efficacy.

CAFs as immune regulators

CAFs modulate both innate and adaptive immune responses through paracrine signalling, direct cell–cell contact, and ECM remodelling.

Suppression of anti-tumour immunity

Multiple CAF subpopulations contribute to immune suppression through distinct mechanisms:

iCAF-mediated immunosuppression: iCAFs secrete a cocktail of immunosuppressive factors including IL-6, CXCL12, CCL2, and prostaglandin E2 (PGE2) (Erve et al., 2024). IL-6 activates STAT3 signalling in CD8+ T cells, impairing their cytotoxic function and promoting exhaustion. CXCL12 recruits regulatory T cells (Tregs) and myeloid-derived suppressor cells (MDSCs), creating an immunosuppressive milieu. CCL2 attracts monocytes that differentiate into tumour-associated macrophages (TAMs) with M2 polarization, further suppressing anti-tumour immunity.

myCAF physical barrier: myCAFs produce a dense, collagen-rich ECM that physically excludes immune cells from tumour cell clusters (Chhabra et al., 2023). This immune-excluded phenotype—characterized by immune cells trapped in the stroma rather than infiltrating tumour nests—is associated with resistance to immune checkpoint inhibitors (ICIs). The mechanical barrier created by myCAFs is reinforced by their contractility, which compresses lymphatic vessels and reduces immune cell trafficking (Fan et al., 2023).

apCAF tolerogenic antigen presentation: apCAFs present tumour antigens to CD4+ T cells via MHC class II without providing adequate co-stimulation, inducing T cell anergy and tolerance (Erve et al., 2024). This mechanism is analogous to peripheral tolerance induction by tolerogenic dendritic cells and may contribute to the failure of anti-tumour immune responses.

Promotion of pro-tumorigenic immunity

Paradoxically, some CAF populations promote inflammatory responses that support tumour growth:

Neutrophil recruitment: iCAF-secreted CXCL1 and CXCL8 attract neutrophils to the tumour site (Acharyya et al., 2012). These tumour-associated neutrophils (TANs) can adopt a pro-tumorigenic N2 phenotype, producing matrix metalloproteinases (MMPs) that degrade the basement membrane and facilitate invasion. TANs also secrete reactive oxygen species (ROS) that promote DNA damage and genomic instability in tumour cells.

Th17 polarization: iCAF-derived IL-6, combined with TGF- β , drives CD4+ T cell differentiation toward the Th17 lineage (Chang et al., 2014). Th17 cells produce IL-17, which promotes tumour angiogenesis and recruits immunosuppressive myeloid cells. In breast cancer, high Th17 infiltration is associated with aggressive disease and poor prognosis.

Macrophage polarization: CAFs promote the differentiation of monocytes into M2-polarized TAMs through secretion of CSF-1, IL-10, and TGF- β (Qian and Pollard, 2011). M2 TAMs suppress anti-tumour immunity, promote angiogenesis, and facilitate metastasis, creating a feed-forward loop that reinforces the immunosuppressive microenvironment.

Immune regulation of CAFs

The crosstalk is bidirectional: immune cells also regulate CAF activation and function.

TGF- β from regulatory T cells: Tregs secrete high levels of TGF- β , which activates fibroblasts and promotes their differentiation into myCAFs (Donnelly et al., 2015). This creates an immunosuppressive niche where Tregs and myCAFs mutually reinforce each other's functions.

Macrophage-derived IL-1 β : M1-polarized macrophages secrete IL-1 β , which activates NF- κ B signalling in fibroblasts, promoting their differentiation into iCAFs (Erve et al., 2024). This mechanism links innate immune activation to CAF heterogeneity, with the inflammatory state of the myeloid compartment determining the balance between myCAF and iCAF subpopulations.

T cell-derived IFN- γ : Activated CD8+ T cells secrete IFN- γ , which can reprogram CAFs toward an anti-tumorigenic phenotype (Kato et al., 2018). IFN- γ -treated CAFs upregulate MHC class I and present tumour antigens, potentially enhancing anti-tumour immunity. However, chronic IFN- γ exposure can also induce CAF senescence and immunosuppressive factor production, highlighting the context-dependent nature of immune–CAF interactions.

Implications for immunotherapy

The CAF–immune crosstalk has direct implications for the efficacy of immunotherapies in breast cancer:

Resistance to immune checkpoint inhibitors: The immune-excluded phenotype created by myCAF capsules is a major mechanism of resistance to anti-PD-1/PD-L1 therapy (Chhabra et al., 2023). Immune cells cannot reach tumour cells to exert their cytotoxic effects, rendering checkpoint blockade ineffective. This explains why breast cancers with high stromal content—particularly TNBC with dense myCAF infiltration—show poor response to immunotherapy despite high tumour mutational burden.

Enhancement of CAR-T cell therapy: CAF-targeted strategies can enhance the efficacy of chimeric antigen receptor T (CAR-T) cell therapy. CAR-T cells engineered to secrete FAP-targeted antibodies can eliminate FAP+ CAFs, disrupting the immunosuppressive niche and improving T cell infiltration (Wang et al., 2014).

Combination strategies: The CAF–immune crosstalk suggests that optimal immunotherapy in breast cancer may require combination approaches that simultaneously target CAFs and immune checkpoints. For example, anti-TGF- β therapy can reduce myCAF activation while enhancing T cell infiltration, potentially synergizing with anti-PD-1 therapy (Chhabra et al., 2023).

CAF-derived extracellular vesicles in immune modulation

CAFs release extracellular vesicles (EVs)—including exosomes and microvesicles—that carry bioactive cargo (proteins, lipids, miRNAs) to immune cells (Zhao et al., 2023). These EVs represent a novel mechanism of CAF–immune communication:

miRNA transfer: CAF-derived exosomes carry miRNAs (miR-21, miR-155, miR-200 family) that reprogram recipient immune cells. miR-21 promotes M2 macrophage polarization, while miR-155 enhances T cell exhaustion (Zhao et al., 2023).

Metabolic reprogramming: CAF-derived EVs carry metabolic enzymes and metabolites (lactate, glutamine) that alter the metabolic state of immune cells, shifting them from oxidative phosphorylation to aerobic glycolysis. This metabolic reprogramming impairs T cell effector function and promotes immune suppression (Zhao et al., 2023).

Antigen transfer: CAF-derived EVs can carry tumour antigens, potentially enabling cross-presentation by dendritic cells and enhancing anti-tumour immunity. However, the net effect of CAF-derived EVs on immune function appears to be immunosuppressive in most contexts, though this remains an active area of investigation.

Therapeutic targeting of CAF–immune crosstalk

Understanding the CAF–immune crosstalk has revealed multiple therapeutic opportunities:

Blocking CAF-derived immunosuppressive factors: Antibodies or small molecules targeting IL-6, CXCL12, or CCL2 can disrupt the immunosuppressive signalling network and enhance anti-tumour immunity (Serve et al., 2023). Clinical trials of IL-6 receptor antibodies (tocilizumab) in combination with ICIs are underway in breast cancer.

Reprogramming CAFs to enhance immunogenicity: Agents that shift CAFs from a suppressive to an immunostimulatory phenotype—such as all-trans retinoic acid (ATRA) or vitamin D—can enhance the efficacy of immunotherapy (Erve et al., 2024). These approaches aim to convert immune-excluded tumours into immune-inflamed tumours.

Disrupting the physical barrier: Strategies that degrade the myCAF-produced ECM—such as hyaluronidase (PEGPH20) or collagenase—can enhance immune cell infiltration into tumour nests (Provenzano et al., 2012). However, these approaches must be carefully balanced to avoid promoting tumour invasion through basement membrane degradation.

Therapeutic Strategies Targeting CAFs

The recognition of CAFs as active contributors to tumour progression has spurred the development of therapeutic strategies aimed at eliminating, reprogramming, or exploiting these cells. This section reviews the current landscape of CAF-targeted therapies in breast cancer, organized by mechanism of action.

CAF depletion strategies

The most direct approach to targeting CAFs is their selective elimination, exploiting surface markers or metabolic dependencies that distinguish them from normal fibroblasts.

FAP-targeted therapies

Fibroblast activation protein (FAP) is a serine protease highly expressed on CAFs but largely absent from normal fibroblasts (Garin-Chesa et al., 2010). This differential expression has made FAP an attractive therapeutic target.

FAP-targeted antibodies: Sibrotuzumab, a humanized anti-FAP antibody, showed modest activity in early-phase clinical trials but failed to demonstrate significant efficacy as monotherapy (Scott et al., 2003). More recently, bispecific antibodies targeting both FAP and immune checkpoints (e.g., FAP × PD-1) have shown enhanced activity in preclinical models by simultaneously depleting CAFs and reactivating T cells (Fang et al., 2023).

FAP-targeted CAR-T cells: CAR-T cells engineered to recognize FAP have shown efficacy in preclinical models of breast cancer, reducing CAF density, disrupting the stromal barrier, and enhancing immune cell infiltration (Wang et al., 2014). However, clinical translation has been limited by on-target, off-tumor toxicity, as FAP is expressed at low levels on normal tissues including bone marrow stromal cells and pancreatic stellate cells.

FAP-targeted radioligand therapy: Radiolabeled FAP inhibitors (FAPIs)—originally developed for diagnostic imaging—have shown therapeutic potential when conjugated to beta-emitting radionuclides (^{90}Y , ^{177}Lu) (Kratovich et al., 2019). In breast cancer, FAPI-based radiotherapy has shown promising responses in early-phase trials, with CAF depletion accompanied by tumour regression and immune activation.

Anti-CAF antibody–drug conjugates

Antibody–drug conjugates (ADCs) targeting CAF-specific antigens represent another depletion strategy. ADCs against podoplanin (PDPN), a glycoprotein expressed on a subset of CAFs, have shown efficacy in preclinical models (Ueki et al., 2023). These agents deliver cytotoxic payloads specifically to CAFs, sparing normal fibroblasts and other cell types.

However, the heterogeneity of CAF marker expression poses a challenge for depletion strategies: no single marker is expressed by all CAF subpopulations, and targeting one marker may spare CAFs expressing alternative phenotypes.

CAF reprogramming strategies

Rather than eliminating CAFs, reprogramming approaches aim to convert tumour-promoting CAFs into quiescent or anti-tumorigenic states.

All-trans retinoic acid

All-trans retinoic acid (ATRA) has been shown to reverse CAF activation in multiple cancer types, including breast cancer (Liu et al., 2023). ATRA inhibits TGF- β -induced SMAD signalling, downregulates α -SMA expression, and reduces ECM production, effectively converting activated myCAF s into quiescent fibroblasts.

In preclinical models, ATRA combined with chemotherapy enhanced drug delivery by reducing stromal density and improving vascular perfusion (Liu et al., 2023). Clinical trials of ATRA in combination with nab-paclitaxel are ongoing in metastatic breast cancer, with preliminary evidence of enhanced anti-tumour activity.

Vitamin D receptor agonists

Vitamin D receptor (VDR) signalling opposes fibroblast activation through inhibition of Hedgehog and TGF- β pathways (Shen et al., 2023). VDR agonists (calcipotriol, paricalcitol) have shown efficacy in reprogramming CAFs in pancreatic cancer, and early evidence suggests similar activity in breast cancer.

In breast cancer models, VDR agonists reduce CAF-derived ECM production, decrease tissue stiffness, and enhance immune cell infiltration (Shen et al., 2023). These effects are accompanied by improved response to chemotherapy and immunotherapy, supporting the concept that CAF reprogramming can sensitize tumours to conventional treatments.

BET inhibitors

Bromodomain and extra-terminal domain (BET) proteins (BRD2/3/4) are epigenetic regulators that maintain the activated CAF phenotype. BET inhibitors (JQ1, I-BET762) disrupt CAF activation by suppressing the transcription of fibrosis-associated genes (Erve et al., 2024).

In breast cancer, BET inhibitors have shown dual effects: they directly suppress tumour cell proliferation while simultaneously reprogramming CAFs toward a quiescent state. This dual activity makes BET inhibitors attractive candidates for combination therapy.

Targeting CAF-derived signalling

Blocking specific signalling pathways downstream of CAFs represents a more targeted approach than global CAF depletion or reprogramming.

CXCR4/CXCL12 axis

The CXCR4/CXCL12 axis mediates multiple CAF functions including immune cell recruitment, angiogenesis, and metastasis. CXCR4 antagonists (plerixafor/AMD3100) block these functions and have shown efficacy in preclinical models of breast cancer (Domanska et al., 2012).

In clinical trials, plerixafor has shown modest single-agent activity but enhanced efficacy when combined with chemotherapy or immunotherapy (Domanska et al., 2012). The combination of plerixafor with anti-PD-1 therapy is being evaluated in clinical trials for metastatic TNBC.

IL-6/STAT3 axis

IL-6, primarily secreted by iCAFs, activates STAT3 signalling in both tumour cells and immune cells, promoting proliferation, survival, and immune evasion. Anti-IL-6 receptor antibodies (tocilizumab) and JAK/STAT3 inhibitors are being evaluated in breast cancer (Chang et al., 2014).

In preclinical models, IL-6 blockade reduces iCAF-mediated immunosuppression and enhances the efficacy of anti-PD-1 therapy (Erve et al., 2024). Clinical trials of tocilizumab in combination with pembrolizumab are underway in metastatic TNBC.

TGF- β signalling

TGF- β is the master regulator of CAF activation and a key mediator of immune suppression. Multiple TGF- β -targeting strategies are in clinical development:

Anti-TGF- β antibodies: Bintrafusp alfa, a bifunctional fusion protein targeting both PD-L1 and TGF- β , showed promising activity in early-phase trials but failed to meet its primary endpoint in a phase

III trial in cervical cancer (Strauss et al., 2023). Ongoing trials are evaluating this agent in breast cancer subtypes with high TGF- β signalling activity.

TGF- β receptor kinase inhibitors: Small-molecule inhibitors of TGF- β receptor I (galunisertib, vactosertib) block CAF activation at the receptor level (Herbert et al., 2015). These agents have shown activity in preclinical models and are being evaluated in early-phase clinical trials.

Anti-TGF- β traps: ATE-Trap, a TGF- β decoy receptor, sequesters TGF- β in the extracellular space, preventing it from activating fibroblasts and immune cells (Teicher, 2021). This approach has shown synergy with anti-PD-1 therapy in preclinical models of breast cancer.

Targeting CAF metabolism

CAFs exhibit distinct metabolic features that can be therapeutically exploited:

Glycolysis inhibition: CAFs rely heavily on aerobic glycolysis (Warburg effect), producing lactate that is exported and consumed by tumour cells (Martinez et al., 2020). Inhibiting glycolysis in CAFs (e.g., with 2-deoxyglucose) reduces lactate production and impairs tumour cell metabolism.

Glutamine metabolism: CAFs are major consumers of glutamine, which they convert to glutamate and other metabolites that support tumour growth (Bertero et al., 2019). Glutaminase inhibitors (CB-839) can block this metabolic support and have shown activity in preclinical breast cancer models.

Lipid metabolism: CAFs exhibit altered lipid metabolism, with increased fatty acid synthesis and oxidation (Sahai et al., 2020). Inhibitors of fatty acid synthase (FASN) or carnitine palmitoyltransferase 1 (CPT1) can disrupt CAF metabolic function and impair their tumour-supportive capabilities.

Combination strategies

Given the complexity of CAF biology, combination approaches targeting multiple CAF functions simultaneously may be more effective than single-agent strategies:

- **CAF depletion + immunotherapy:** Combining FAP-targeted CAR-T cells with anti-PD-1 therapy has shown enhanced efficacy in preclinical models by simultaneously removing the physical barrier and reactivating T cells.
- **CAF reprogramming + chemotherapy:** ATRA combined with nab-paclitaxel enhances drug delivery by reducing stromal density while simultaneously reducing CAF-mediated chemoresistance.
- **Anti-TGF- β + anti-PD-1:** Blocking TGF- β signalling reduces both CAF activation and immune suppression, potentially converting immune-excluded tumours into immune-inflamed tumours.
- **Metabolic inhibitors + immunotherapy:** Targeting CAF metabolism can reduce immunosuppressive metabolite production, enhancing the efficacy of immune checkpoint blockade.

These combination strategies reflect the understanding that CAFs contribute to tumour progression through multiple, redundant mechanisms, and that effective therapy must address this complexity.

CAFs as Mediators of Treatment Resistance

A growing body of evidence indicates that CAFs contribute to therapeutic resistance in breast cancer through multiple mechanisms. Understanding these mechanisms is essential for developing strategies to overcome resistance and improve treatment outcomes.

Chemoresistance

CAFs promote resistance to chemotherapy through several interconnected pathways:

Physical barrier and drug delivery

The dense, collagen-rich ECM produced by myCAFs creates a physical barrier that impedes drug penetration into tumour tissue (Provenzano et al., 2012). This barrier increases interstitial fluid pressure, compresses blood vessels, and reduces perfusion, limiting the delivery of systemically administered chemotherapeutics.

In breast cancer, high stromal density—measured by collagen content or imaging features—is associated with poor response to neoadjuvant chemotherapy (Farinha et al., 2019). Preclinical studies have shown that strategies to reduce stromal density (e.g., anti-TGF- β therapy, ATRA) can enhance drug delivery and improve chemosensitivity (Liu et al., 2023).

Paracrine survival signalling

CAFs secrete multiple factors that directly promote tumour cell survival in the presence of chemotherapy:

IL-6: CAF-derived IL-6 activates STAT3 signalling in tumour cells, upregulating anti-apoptotic proteins (BCL-2, MCL-1) and drug efflux pumps (ABCB1, ABCG2) (Chang et al., 2014). This paracrine survival signalling is a major mechanism of resistance to taxanes and anthracyclines in breast cancer.

CXCL12: CAF-secreted CXCL12 activates CXCR4 on tumour cells, triggering PI3K/AKT signalling that promotes survival and proliferation (Orimo et al., 2005). Inhibition of CXCR4 with plerixafor sensitizes breast cancer cells to chemotherapy in preclinical models.

HGF: Hepatocyte growth factor (HGF) secreted by CAFs activates MET signalling in tumour cells, promoting survival, invasion, and drug resistance (Wilson et al., 2014). HGF/MET signalling is particularly relevant in basal-like breast cancer, where it contributes to resistance to EGFR-targeted therapies.

Exosome-mediated resistance

CAFs release exosomes containing miRNAs and proteins that confer chemoresistance to tumour cells (Zhao et al., 2023):

miR-21: CAF-derived exosomes carry miR-21, which downregulates the pro-apoptotic protein PDCD4, enhancing tumour cell survival during chemotherapy (Zhao et al., 2023).

miR-200 family: Paradoxically, CAF-derived exosomes can also carry miR-200 family members that suppress EMT and enhance chemosensitivity, highlighting the complex and context-dependent nature of exosome-mediated communication.

Drug efflux pumps: CAF-derived exosomes can transfer functional drug efflux pumps (P-glycoprotein) to tumour cells, directly conferring multidrug resistance (Zhao et al., 2023).

Endocrine resistance

In ER+ breast cancer, CAFs contribute to resistance to endocrine therapies (tamoxifen, aromatase inhibitors) through multiple mechanisms:

Alternative growth factor signalling

CAFs secrete growth factors (FGF, HGF, IGF) that activate signalling pathways parallel to ER, providing tumour cells with ER-independent survival signals (Seth et al., 2021). These alternative pathways bypass the requirement for estrogen signalling, rendering endocrine therapy ineffective.

In particular, CAF-derived FGF2 activates FGFR signalling in tumour cells, promoting proliferation and survival in the absence of estrogen (Seth et al., 2021). FGFR inhibitors (erdafitinib, pemigatinib) are being evaluated in combination with endocrine therapy for ER+ breast cancer with CAF-mediated resistance.

ECM-mediated resistance

The ECM produced by CAFs provides physical and biochemical support that enhances tumour cell survival during endocrine therapy. Integrin signalling from ECM components activates FAK and PI3K/AKT pathways, promoting resistance to tamoxifen-induced apoptosis (Seth et al., 2021).

Immunotherapy resistance

CAFs are major contributors to resistance to immune checkpoint inhibitors in breast cancer:

Immune exclusion

As discussed in Section 5, myCAFs create a physical barrier that excludes immune cells from tumour nests, preventing the anti-tumour immune response required for ICI efficacy (Chhabra et al., 2023). This immune-excluded phenotype is particularly common in TNBC with dense stroma, explaining the poor response to anti-PD-1/PD-L1 therapy in these tumours.

Immunosuppressive milieu

iCAFs create an immunosuppressive microenvironment through secretion of IL-6, CXCL12, and CCL2, which recruit and activate immunosuppressive cells (Tregs, MDSCs, M2 macrophages) (Erve et al., 2024). This immunosuppressive milieu impairs T cell function and limits the efficacy of ICIs.

Antigen presentation without co-stimulation

apCAFs present tumour antigens to CD4+ T cells via MHC class II without adequate co-stimulation, inducing T cell anergy and tolerance (Erve et al., 2024). This tolerogenic mechanism may contribute to the failure of anti-tumour immune responses despite ICI therapy.

Targeted therapy resistance

CAFs also contribute to resistance to targeted therapies in breast cancer:

HER2+ breast cancer: CAFs secrete neuregulin (NRG1) and other HER3 ligands that activate HER2/HER3 heterodimers, providing an alternative survival pathway when HER2 is blocked by trastuzumab or lapatinib (Seth et al., 2021).

CDK4/6 inhibitor resistance: CAF-derived FGF activates FGFR signalling, which can bypass CDK4/6 inhibition by activating alternative cell cycle regulators (Seth et al., 2021).

PI3K inhibitor resistance: CAFs secrete insulin-like growth factor (IGF) that activates PI3K signalling through IGF1R, providing an alternative activation pathway when PI3K is inhibited (Seth et al., 2021).

Clinical implications of CAF-mediated resistance

Understanding CAF-mediated resistance has important clinical implications:

Biomarker development: Stromal features—including CAF density, collagen organization, and CAF subpopulation composition—may serve as predictive biomarkers for treatment response. High myCAF density, for example, may predict poor response to immunotherapy.

Combination therapy: Strategies that target CAFs in combination with conventional therapies may overcome resistance. For example, combining anti-TGF- β therapy with anti-PD-1 may convert immune-excluded tumours into immune-inflamed tumours, enhancing ICI efficacy.

Sequential therapy: Understanding the temporal dynamics of CAF-mediated resistance may guide the sequencing of therapies. For example, CAF-targeted therapy prior to chemotherapy may enhance drug delivery and improve response.

CAFs as Biomarkers in Breast Cancer

The heterogeneity and functional diversity of CAFs suggest their potential as biomarkers for diagnosis, prognosis, and treatment response prediction in breast cancer.

Prognostic biomarkers

Multiple CAF-related features have been associated with clinical outcomes in breast cancer:

CAF density and distribution

High overall CAF density—measured by α -SMA or FAP staining—is associated with poor prognosis in most breast cancer subtypes (Erve et al., 2024). However, the prognostic significance varies by CAF subpopulation:

myCAF density: High myCAF density is associated with aggressive disease features including large tumour size, high grade, and lymph node metastasis (Erve et al., 2024). myCAFs contribute to tumour progression through ECM production, mechanical signalling, and immune exclusion, explaining their association with poor outcomes.

iCAF density: The prognostic significance of iCAFs is context-dependent. In some studies, high iCAF density is associated with poor prognosis, reflecting the pro-tumorigenic functions of inflammatory cytokines (Erve et al., 2024). In other studies, iCAF density correlates with enhanced immune infiltration and better response to immunotherapy, suggesting a more complex relationship.

apCAF density: Limited data exist on the prognostic significance of apCAFs. Preliminary studies suggest that high apCAF density may be associated with better prognosis in ER+ breast cancer, potentially reflecting enhanced anti-tumour immune responses (Erve et al., 2024).

Spatial organization

Beyond density, the spatial organization of CAFs provides prognostic information:

CAF–tumour distance: Short distance between CAFs and tumour cells is associated with poor prognosis, reflecting the importance of paracrine signalling in tumour promotion (Erve et al., 2024).

CAF clustering: Highly clustered CAF distributions—indicating organized stromal niches—are associated with aggressive disease, while dispersed CAF distributions are associated with better outcomes (Erve et al., 2024).

Immune–CAF spatial relationships: The spatial relationship between CAFs and immune cells provides prognostic information. Tumours with CAFs excluded from immune cell-rich areas (immune-excluded phenotype) have worse outcomes than tumours with CAFs integrated within immune infiltrates (immune-infiltrated phenotype) (Chhabra et al., 2023).

Predictive biomarkers

CAF features may predict response to specific therapies:

Immunotherapy response prediction

The immune-excluded phenotype—characterized by high myCAF density surrounding tumour cell clusters—is associated with poor response to immune checkpoint inhibitors (Chhabra et al., 2023). This phenotype can be identified by spatial analysis of α -SMA+ myCAFs relative to CD8+ T cells.

Conversely, tumours with high iCAF density and dispersed CAF distributions (immune-inflamed phenotype) show better response to immunotherapy (Chhabra et al., 2023). This suggests that CAF subtyping and spatial analysis may help identify patients most likely to benefit from ICI therapy.

Chemotherapy response prediction

CAF-related features may predict response to chemotherapy:

Stromal gene signatures: Gene expression signatures derived from CAFs—including ECM components, TGF- β targets, and inflammatory cytokines—have been associated with resistance to neoadjuvant chemotherapy in breast cancer (Farinha et al., 2019).

Collagen organization: The organization of collagen fibres—measured by second harmonic generation imaging or spatial transcriptomics—is associated with chemotherapy response. Tumours with aligned, dense collagen (reflecting high myCAF activity) show poor response to chemotherapy (Provenzano et al., 2012).

CAF density: High overall CAF density is associated with resistance to neoadjuvant chemotherapy, reflecting the physical barrier and paracrine survival signalling provided by CAFs (Farinha et al., 2019).

Circulating CAF biomarkers

Circulating CAFs and CAF-derived factors represent a minimally invasive source of biomarkers:

Circulating fibroblasts

Circulating fibroblast-like cells—termed circulating fibrocytes—have been detected in the peripheral blood of breast cancer patients (Abe et al., 2012). High circulating fibrocyte counts are associated

with advanced disease and poor prognosis, potentially reflecting the systemic mobilization of CAF precursors.

However, the detection and characterization of circulating fibrocytes remains technically challenging, and their clinical utility as biomarkers requires further validation.

Circulating CAF-derived factors

CAFs secrete multiple factors that can be measured in peripheral blood:

CA125 (MUC16): While traditionally associated with ovarian cancer, CA125 is also secreted by CAFs in breast cancer (Seth et al., 2021). Elevated serum CA125 levels are associated with advanced disease and poor prognosis.

Tenascin-C: This ECM glycoprotein, produced by CAFs, is elevated in the serum of breast cancer patients and correlates with disease stage and metastatic burden (Seth et al., 2021).

Periostin: This matricellular protein, secreted by myCAFs, is elevated in the serum of breast cancer patients and is associated with poor prognosis (Seth et al., 2021).

Circulating exosomes

CAFs release exosomes into the bloodstream that carry CAF-specific cargo (miRNAs, proteins) (Zhao et al., 2023). These circulating exosomes can be isolated and characterized, providing information about CAF activity without the need for tissue biopsy.

CAF-derived exosomal miRNAs—including miR-21, miR-155, and miR-200 family members—are elevated in the plasma of breast cancer patients and correlate with disease stage and prognosis (Zhao et al., 2023).

Imaging-based CAF biomarkers

Non-invasive imaging techniques can capture CAF-related features:

MRI-based stromal features

Magnetic resonance imaging (MRI) can assess stromal features that reflect CAF activity:

Apparent diffusion coefficient (ADC): Low ADC values—indicating restricted water diffusion—correspond to dense, collagen-rich stroma produced by myCAFs. Low ADC is associated with poor prognosis and chemoresistance in breast cancer (Seth et al., 2021).

Dynamic contrast enhancement (DCE): The pattern of contrast enhancement reflects vascular perfusion, which is influenced by CAF-mediated vascular remodelling. Rapid wash-in and wash-out patterns are associated with high CAF activity and poor prognosis (Seth et al., 2021).

PET-based CAF imaging

Positron emission tomography (PET) with FAP-targeted tracers (^{68}Ga -FAPI) can visualize CAF-rich stromal compartments (Kratochwil et al., 2019). High FAPI uptake is associated with dense stroma and may predict resistance to immunotherapy.

Multi-modal CAF biomarker panels

The heterogeneity of CAFs suggests that single biomarkers may be insufficient to capture their full biological complexity. Multi-modal biomarker panels—integrating CAF density, subpopulation composition, spatial organization, and circulating factors—may provide more accurate prognostic and predictive information.

Preliminary studies have shown that combining CAF-related features with tumour-intrinsic features (gene expression, mutation status) improves the prediction of treatment response and clinical outcomes (Seth et al., 2021). However, the development and validation of such multi-modal panels requires large, well-annotated clinical cohorts with comprehensive molecular and spatial data.

Future Directions

Despite significant advances in our understanding of CAF biology in breast cancer, critical knowledge gaps remain. We identify six directions likely to shape future research and clinical translation.

Foundation models for CAF biology

Large-scale pre-trained models—analogueous to those emerging for spatial omics (Nicheformer, scGPT)—could transform CAF research by learning transferable representations of CAF identity and function from atlas-scale data (Cui et al., 2024). These models could:

- Enable automated CAF subtyping across diverse breast cancer datasets, reducing the need for manual annotation.
- Predict CAF functional states from limited transcriptomic data, facilitating the analysis of clinical samples with sparse molecular information.
- Identify conserved CAF programmes across cancer types, revealing fundamental principles of CAF biology.

However, the development of CAF-specific foundation models requires large, standardized training datasets with comprehensive CAF annotations—a resource that does not yet exist. Community efforts to generate such datasets, analogueous to the Human Cell Atlas, would be transformative.

Spatial multi-omics integration

The integration of spatial transcriptomics, spatial proteomics, and spatial metabolomics will provide a comprehensive view of CAF biology at single-cell resolution (Ståhl et al., 2016):

Spatial transcriptomics + proteomics: Combining gene expression with protein abundance will enable the mapping of CAF signalling pathways at spatial resolution, revealing how transcriptional programmes translate into functional outputs.

Spatial transcriptomics + metabolomics: Spatial metabolomics technologies (MALDI-MSI, DESI-MSI) can map metabolite distributions within tissues. Integrating these data with CAF transcriptomics will reveal how CAF metabolism shapes the metabolic landscape of the tumour microenvironment.

Spatial multi-modal atlases: Comprehensive spatial atlases integrating transcriptomic, proteomic, and metabolomic data from the same tissue sections will provide unprecedented resolution of CAF biology. The Human Tumour Atlas Network (HTAN) and similar initiatives are beginning to generate such resources.

Functional genomics of CAFs

CRISPR-based functional genomics screens—performed directly in CAFs within the tumour microenvironment—could identify the genetic dependencies that maintain the CAF phenotype and drive their tumour-promoting functions:

In vivo CRISPR screens: Performing CRISPR knockout screens in CAFs within orthotopic breast tumour models could identify genes essential for CAF activation, survival, and function (Erve et al., 2024).

Single-cell CRISPR screens: Combining CRISPR perturbation with single-cell readouts (CROP-seq, Perturb-seq) could reveal the gene regulatory networks that control CAF heterogeneity and function.

CRISPRi/CRISPRa screens: Interference and activation screens could identify genes whose modulation reprograms CAFs from tumour-promoting to tumour-restraining states, revealing therapeutic targets for CAF reprogramming.

CAF-targeted drug delivery

The development of CAF-targeted drug delivery systems could enhance the efficacy of CAF-targeted therapies while reducing off-target toxicity:

Nanoparticle-based delivery: Nanoparticles functionalized with FAP-targeting peptides or antibodies can deliver drugs specifically to CAFs, reducing systemic exposure (Miao et al., 2023).

Exosome-based delivery: Engineered exosomes carrying therapeutic cargo (siRNAs, small molecules) can be targeted to CAFs through surface modifications, exploiting the natural tropism of exosomes for the tumour microenvironment (Zhao et al., 2023).

CAR-T cell-mediated delivery: CAR-T cells engineered to secrete therapeutic payloads upon CAF recognition can deliver drugs directly to the CAF-rich stroma, combining CAF depletion with drug delivery.

Liquid biopsy for CAF monitoring

The development of liquid biopsy assays for CAF-derived biomarkers could enable real-time monitoring of CAF activity and treatment response:

CAF-derived exosomal miRNAs: Standardized assays for CAF-specific exosomal miRNAs could provide minimally invasive monitoring of CAF activity during treatment (Zhao et al., 2023).

Circulating CAF proteins: Multiplexed immunoassays for CAF-secreted proteins (tenascin-C, periostin, FGF2) could track CAF activity over time and predict treatment resistance.

CAF-derived ctDNA: While CAFs do not harbour tumour-specific mutations, they may release cell-free DNA with distinct epigenetic signatures (methylation patterns) that could serve as CAF-specific biomarkers.

Clinical trial design for CAF-targeted therapies

The clinical development of CAF-targeted therapies requires innovative trial designs:

Biomarker-driven patient selection: Trials should incorporate CAF-related biomarkers (CAF density, subpopulation composition, spatial organization) to identify patients most likely to benefit from CAF-targeted therapy.

Adaptive trial designs: Given the heterogeneity of CAF biology, adaptive trial designs that allow treatment modifications based on CAF biomarker changes during therapy may be more efficient than fixed designs.

Combination therapy platforms: Platform trials evaluating multiple CAF-targeted strategies in combination with standard therapies could accelerate the identification of effective combinations.

Translational endpoints: Trials should incorporate translational endpoints—including pre- and post-treatment biopsies with single-cell and spatial analyses—to understand the mechanisms of response and resistance.

Ethical considerations

As CAF-targeted therapies move toward clinical translation, ethical considerations must be addressed:

Equity in access: CAF-targeted therapies—particularly CAR-T cells and bispecific antibodies—are likely to be expensive. Ensuring equitable access to these therapies across diverse populations is essential.

Long-term safety: The long-term consequences of depleting or reprogramming CAFs—which play roles in wound healing and tissue homeostasis—must be carefully monitored.

Informed consent: Patients enrolled in CAF-targeted trials must be informed about the experimental nature of these approaches and the potential for unexpected toxicities.

Conclusion

Cancer-associated fibroblasts have emerged from the shadows of the tumour microenvironment to become a central focus of breast cancer research. The past decade—particularly the era of single-cell and spatial omics—has transformed our understanding of CAF biology, revealing a degree of heterogeneity and functional diversity that was unimaginable when these cells were first described as “activated fibroblasts.”

We have organized this review around five interconnected themes that capture the current state of CAF research in breast cancer. Seven key messages emerge:

1. **CAFs are heterogeneous, not homogeneous.** The myCAF/iCAF/apCAF classification provides a useful framework, but the true CAF landscape is far more complex, with multiple subpopulations and continuous transcriptional states shaped by local signalling gradients, ECM composition, and immune context.
2. **CAF origins matter.** CAFs can arise from tissue-resident fibroblasts, bone marrow-derived mesenchymal stem cells, endothelial cells, and even tumour cells through EMT. These different origins may confer distinct functional properties and therapeutic vulnerabilities.
3. **Spatial organization determines function.** The spatial distribution of CAF subpopulations—their positioning relative to tumour cells, immune cells, and vasculature—is a critical determinant of their biological effects. Spatial transcriptomics is revealing organized CAF niches that were invisible to bulk and dissociated single-cell analyses.
4. **CAF-immune crosstalk is bidirectional and therapeutically actionable.** CAFs shape the immune landscape through paracrine signalling, physical barrier formation, and tolerogenic antigen presentation. Conversely, immune cells regulate CAF activation and heterogeneity. This crosstalk is a major determinant of immunotherapy efficacy.

5. **CAFs are mediators of treatment resistance.** CAFs contribute to resistance to chemotherapy, endocrine therapy, immunotherapy, and targeted therapy through multiple mechanisms. Understanding these mechanisms is essential for developing combination strategies that overcome resistance.
6. **CAF-targeted therapies are entering clinical trials.** Multiple strategies—including FAP-targeted antibodies, CAF reprogramming agents, and signalling pathway inhibitors—are being evaluated in clinical trials. The success of these approaches will depend on biomarker-driven patient selection and rational combination strategies.
7. **Critical knowledge gaps remain.** The field lacks standardized CAF nomenclature, validated CAF-specific biomarkers, and prospective clinical trials demonstrating the efficacy of CAF-targeted therapies. Addressing these gaps requires community-wide coordination and investment.

As spatial omics technologies mature and functional genomics approaches are applied to CAFs, we anticipate a new wave of discoveries that will further refine our understanding of CAF biology and open new therapeutic opportunities. The ultimate goal—translating CAF biology into improved outcomes for breast cancer patients—remains within reach, but will require the continued integration of basic, translational, and clinical research.

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